Day-4

```
mysql> USE PFS7;
Database changed
mysql> CREATE TABLE EMPLOYEES(
    -> EMPID CHAR(5),
    -> FNAME VARCHAR(100),
    -> LNAME VARCHAR(60),
    -> AGE INT,
    -> DOJ DATE,
    -> ADDRESS TINYTEXT,
    -> DEPT VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.30 sec)

mysql> DESC EMPLOYEES;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| EMPID   | char(5)     | YES  |     | NULL    |       |
| FNAME   | varchar(100) | YES  |     | NULL    |       |
| LNAME   | varchar(60)  | YES  |     | NULL    |       |
| AGE     | int         | YES  |     | NULL    |       |
| DOJ     | date        | YES  |     | NULL    |       |
| ADDRESS | tinytext    | YES  |     | NULL    |       |
| DEPT    | varchar(50)  | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
7 rows in set (0.04 sec)

mysql> -- ALTER
mysql> -- MODIFY THE FNAME COLUMN IN EMPLOYEES TABLE TO
VARCHAR(70)
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME MODIFY COLUMN-NAME
DATATYPE SIZE.
mysql> ALTER TABLE EMPLOYEES MODIFY FNAME VARCHAR(70);
Query OK, 0 rows affected (0.34 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC EMPLOYEES;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| EMPID   | char(5)     | YES  |     | NULL    |       |
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```
| FNAME   | varchar(70) | YES  |     | NULL    |       |
| LNAME   | varchar(60) | YES  |     | NULL    |       |
| AGE     | int         | YES  |     | NULL    |       |
| DOJ     | date        | YES  |     | NULL    |       |
| ADDRESS | tinytext    | YES  |     | NULL    |       |
| DEPT    | varchar(50) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)

mysql> ALTER TABLE EMPLOYEES MODIFY LNAME VARCHAR(80);
Query OK, 0 rows affected (2.27 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC EMPLOYEES;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| EMPID   | char(5)     | YES  |     | NULL    |       |
| FNAME   | varchar(70) | YES  |     | NULL    |       |
| LNAME   | varchar(80) | YES  |     | NULL    |       |
| AGE     | int         | YES  |     | NULL    |       |
| DOJ     | date        | YES  |     | NULL    |       |
| ADDRESS | tinytext    | YES  |     | NULL    |       |
| DEPT    | varchar(50) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)

mysql> -- ADD A NEW COLUMN NAAMED WITH LOCATION OF TYPE
TINYTEXT TO THE EMPLOYEES TABLE.
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME ADD COLUMN-NAME
DATATYPE(SIZE)
mysql> ALTER TABLE EMPLOYEE ADD LOCATION TINYTEXT;
ERROR 1146 (42S02): Table 'pfs3.employee' doesn't exist
mysql> ALTER TABLE EMPLOYEES ADD LOCATION TINYTEXT;
Query OK, 0 rows affected (0.34 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC EMPLOYEES;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| EMPID    | char(5)     | YES  |     | NULL    |       |
```

```
| FNAME    | varchar(70) | YES  |     | NULL    |       |
| LNAME    | varchar(80) | YES  |     | NULL    |       |
| AGE      | int         | YES  |     | NULL    |       |
| DOJ      | date        | YES  |     | NULL    |       |
| ADDRESS  | tinytext    | YES  |     | NULL    |       |
| DEPT     | varchar(50) | YES  |     | NULL    |       |
| LOCATION | tinytext    | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
8 rows in set (0.04 sec)

mysql> -- TO ADD A COLUMN AFTER A PARTICULAR COLUMN
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME ADD COLUMN-NAME
AFTER SPECIFIED-COLUMN NAME
mysql> ALTER TABLE EMPLOYEES ADD PFID INT AFTER ADDRESS;
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC EMPLOYEES;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| EMPID    | char(5)     | YES  |     | NULL    |       |
| FNAME    | varchar(70) | YES  |     | NULL    |       |
| LNAME    | varchar(80) | YES  |     | NULL    |       |
| AGE      | int         | YES  |     | NULL    |       |
| DOJ      | date        | YES  |     | NULL    |       |
| ADDRESS  | tinytext    | YES  |     | NULL    |       |
| PFID     | int         | YES  |     | NULL    |       |
| DEPT     | varchar(50) | YES  |     | NULL    |       |
| LOCATION | tinytext    | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
9 rows in set (0.02 sec)

mysql> -- RENAME THE FNAME COLUMN TO FIRSTNAME IN THE
EMPLOYEES TABLE;
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME CHANGE
EXISTING-COLUMN-NAME NEW COLUMN-NAME DATATYPE(SIZE)
mysql> ALTER TABLE EMPLOYEES CHANGE COLUMN FNAME FIRSTNAME
VARCHAR(30);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESC EMPLOYEES;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| EMPID     | char(5)     | YES  |     | NULL    |       |
| FIRSTNAME | varchar(30) | YES  |     | NULL    |       |
| LNAME     | varchar(80) | YES  |     | NULL    |       |
| AGE       | int         | YES  |     | NULL    |       |
| DOJ       | date        | YES  |     | NULL    |       |
| ADDRESS   | tinytext    | YES  |     | NULL    |       |
| PFID      | int         | YES  |     | NULL    |       |
| DEPT      | varchar(50) | YES  |     | NULL    |       |
| LOCATION  | tinytext    | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
9 rows in set (0.00 sec)

mysql> ALTER TABLE EMPLOYEES CHANGE COLUMN LNAME LASTNAME
VARCHAR(30);
Query OK, 0 rows affected (0.20 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC EMPLOYEES;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| EMPID     | char(5)     | YES  |     | NULL    |       |
| FIRSTNAME | varchar(30) | YES  |     | NULL    |       |
| LASTNAME  | varchar(30) | YES  |     | NULL    |       |
| AGE       | int         | YES  |     | NULL    |       |
| DOJ       | date        | YES  |     | NULL    |       |
| ADDRESS   | tinytext    | YES  |     | NULL    |       |
| PFID      | int         | YES  |     | NULL    |       |
| DEPT      | varchar(50) | YES  |     | NULL    |       |
| LOCATION  | tinytext    | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
9 rows in set (0.01 sec)

mysql> -- HOW TO DROP A COLUMN
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME DROP COLUMN
COLUMN-NAME
mysql> -- DROP THE COLUMN PFID IN EMPLOYEES TABLE
mysql> ALTER TABLE EMPLOYEES DROP COLUMN PFI;
```

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ERROR 1091 (42000): Can't DROP 'PFI'; check that column/key exists
mysql> ALTER TABLE EMPLOYEES DROP COLUMN PFID;
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC EMPLOYEES;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| EMPID     | char(5)     | YES  |     | NULL    |       |
| FIRSTNAME | varchar(30) | YES  |     | NULL    |       |
| LASTNAME  | varchar(30) | YES  |     | NULL    |       |
| AGE       | int         | YES  |     | NULL    |       |
| DOJ       | date        | YES  |     | NULL    |       |
| ADDRESS   | tinytext    | YES  |     | NULL    |       |
| DEPT      | varchar(50) | YES  |     | NULL    |       |
| LOCATION  | tinytext    | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
8 rows in set (0.03 sec)

mysql> ALTER TABLE EMPLOYEES DROP COLUMN LOCATION;
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC EMPLOYEES;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| EMPID     | char(5)     | YES  |     | NULL    |       |
| FIRSTNAME | varchar(30) | YES  |     | NULL    |       |
| LASTNAME  | varchar(30) | YES  |     | NULL    |       |
| AGE       | int         | YES  |     | NULL    |       |
| DOJ       | date        | YES  |     | NULL    |       |
| ADDRESS   | tinytext    | YES  |     | NULL    |       |
| DEPT      | varchar(50) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
7 rows in set (0.00 sec)

mysql> -- HOW TO CHANGE TABLE -NAME
mysql> -- SYNTAX : ALTER TABLE TABLE-NAME RENAME TO
NEW-TABLE-NAME
```

```
mysql> -- CHANGE THE TABLE NAME EMPLOYEES TO CODEGNAN_EMP
TABLE
mysql> ALTER TABLE EMPLOYEES RENAME TO CODEGNAN_EMP;
Query OK, 0 rows affected (0.06 sec)

mysql> DESC EMPLOYEES;
ERROR 1146 (42S02): Table 'pfs3.employees' doesn't exist
mysql> SHOW TABLES;
+---------------+
| Tables_in_pfs3 |
+---------------+
| codegnan_emp   |
+---------------+
1 row in set (0.03 sec)

mysql> DESC CODEGNAN_EMP;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| EMPID     | char(5)     | YES  |     | NULL    |       |
| FIRSTNAME | varchar(30) | YES  |     | NULL    |       |
| LASTNAME  | varchar(30) | YES  |     | NULL    |       |
| AGE       | int         | YES  |     | NULL    |       |
| DOJ       | date        | YES  |     | NULL    |       |
| ADDRESS   | tinytext    | YES  |     | NULL    |       |
| DEPT      | varchar(50) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
7 rows in set (0.00 sec)

mysql> ALTER TABLE CODEGNAN_EMP RENAME TO EMPLOYEES;
Query OK, 0 rows affected (0.07 sec)

mysql> SHOW TABLES;
+---------------+
| Tables_in_pfs3 |
+---------------+
| employees      |
+---------------+
1 row in set (0.01 sec)

mysql> NOTE
```

Part A: Create Table Assignment
Create a table named company_staff.
Add the following columns to the table:
emp_id, age, experience, gender, emp_name, address, salary, department, skills, joining_date.
Use INT datatype for emp_id.
Use TINYINT datatype for age.
Use SMALLINT datatype for experience.
Use CHAR(1) datatype for gender.
Use VARCHAR(50) datatype for emp_name.
Use TINYTEXT datatype for address.
Use DECIMAL(10,2) datatype for salary.
Use ENUM('HR','IT','Sales','Finance','Support') datatype for the department.
Use SET('Java','Python','SQL','Excel','Communication') datatype for skills.
Use DATE datatype for joining_date.
Make emp_id as PRIMARY KEY.
Display the structure of the table after creation.
Show all tables available in the database.

```
CREATE TABLE company_staff (
    emp_id INT primary key ,
    age TINYINT,
    experience SMALLINT,
    gender CHAR(1),
    emp_name VARCHAR(50),
    address TINYTEXT,
    salary DECIMAL(10,2),
    department ENUM('HR','IT','Sales','Finance','Support'),
    skills SET('Java','Python','SQL','Excel','Communication'),
    joining_date DATE
);
```